MARKING SCHEME

Model Paper — 01

Third Term Test

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Note (for teachers):

- Part I – A: 2 marks per question (normally 1 mark for the correct method and 1 mark for the correct final answer). — Total 30 marks
- Part I – B: 10 marks per question. — Total 20 marks. The marks for each step are shown in the right-hand column.
- Part II: candidates answer any five questions only; 10 marks per question. — Total 50 marks. If more than five are answered, mark the first five attempted.
- Award full marks for any correct alternative method. Accept a reasonable range for values read from a graph.

Part I – A — Answers

1. Common difference = 3 ; $T_n = 5 + (n - 1) \times 3$

Final answer: $3n + 2$

(2 marks)

2. $45 = 32 + 8 + 4 + 1 = 1\times2^5 + 0\times2^4 + 1\times2^3 + 1\times2^2 + 0\times2 + 1$ (or by repeated division by 2: remainders 1, 0, 1, 1, 0, 1 read from bottom to top)

Final answer: 101101_2

(2 marks)

3. $\frac{7}{3} \div \frac{7}{4} = \frac{7}{3} \times \frac{4}{7}$

Final answer: $\frac{4}{3} = 1\frac{1}{3}$

(2 marks)

4. Discount = $2400 \times \frac{15}{100} = 360$; selling price = $2400 - 360$

Final answer: Rs 2040

(2 marks)

5. The two numbers with sum 7 and product 12 are 3 and 4

Final answer: $(x + 3)(x + 4)$

(2 marks)

6. $a^{3+5-(-2)}$

Final answer: a^{10}

(2 marks)

7. $0.000507 = 5.07 \times 10^{-4}$ (decimal point moved 4 places to the right)

Final answer: 5.07×10^{-4}

(2 marks)

8. $2x - 6 = x + 4 \Rightarrow 2x - x = 4 + 6$ (check: $2(10 - 3) = 14 = 10 + 4$)

Final answer: $x = 10$

(2 marks)

9. Since AB ∥ CD, 65° and x are co-interior angles ; $x = 180° - 65°$

Final answer: 115°

(2 marks)

10. Exterior angle = sum of the two interior opposite angles ; $120° = 50° + x$

Final answer: $x = 70°$

(2 marks)

11. $2\pi r = 2 \times \frac{22}{7} \times 7$

Final answer: 44 cm

(2 marks)

12. $AC^2 = 5^2 + 12^2 = 25 + 144 = 169$ (Pythagoras theorem)

Final answer: AC = 13 cm

(2 marks)

13. $2x < 11 - 3 \Rightarrow 2x < 8 \Rightarrow x < 4$

Final answer: $x < 4$; greatest integer value 3

(2 marks)

14. A = {1, 2, 3, 4, 6, 12} ; B = {1, 2, 3, 6, 9, 18}

Final answer: $A \cap B = \{1, 2, 3, 6\}$

(2 marks)

15. Number of red balls ÷ total number of balls = $\frac{3}{3+5}$

Final answer: $\frac{3}{8}$

(2 marks)

Part I – B — Answers and mark allocation

Question 1

(i)	
Area = length × breadth = 20 × 14	**1**
= 280 m^2	**1**
(ii)	
$\pi r^2 = \frac{22}{7} \times 3.5 \times 3.5$	**1**
= 38.5 m^2	**1**
(iii)	
Area of garden excluding the pond = 280 − 38.5	**1**
= 241.5 m^2	**1**
(iv)	
$2\pi r = 2 \times \frac{22}{7} \times 3.5$	**1**
= 22 m	**1**
(v)	
Cost = 22 × 150	**1**
= Rs 3300	**1**

Total marks: 10

Question 2

(i)	
A rectangle for the universal set ξ containing two intersecting circles, correctly labelled C and V	**1**
7 in the intersection ($C \cap V$) ; 25 − 7 = 18 in the C-only region	**1**
18 − 7 = 11 in the V-only region ; 4 (= 40 − 36) outside both circles	**1**
(ii)	
$n(C \cup V) = 25 + 18 - 7 = 36$ (or 18 + 7 + 11)	**1**
Students who play neither game = 40 − 36 = 4	**1**
(iii)	
Cricket only = 25 − 7	**1**
= 18	**1**
(iv)	
Volleyball only = 18 − 7 = 11	**1**
Probability = number of favourable outcomes ÷ total number of students = 11 ÷ 40	**1**
$P = \frac{11}{40}$	**1**

Total marks: 10

Part II — Answers and mark allocation

Question 1

(a)(i)	
Marked price = $48\,000 \times \frac{125}{100}$	**1**
= Rs 60 000	**1**
(a)(ii)	
Selling price = $60\,000 \times \frac{90}{100}$	**1**
= Rs 54 000	**1**
(a)(iii)	
Profit = 54 000 − 48 000 = Rs 6 000	**1**
Profit percentage = $\frac{6\,000}{48\,000} \times 100$	**1**
= 12.5%	**1**
(b)	
Price of 1 kg of rice = 1150 ÷ 5 = Rs 230	**1**
Price of 8 kg of rice = 230 × 8	**1**
= Rs 1840	**1**

Total marks: 10

Question 2

(a)	
(i) $2x^2 - 8 = 2(x^2 - 4)$	**1**
$= 2(x - 2)(x + 2)$	**1**
(ii) $x^2 - 5x - 14$: the numbers with product -14 and sum -5 are -7 and $+2$	**1**
$= (x - 7)(x + 2)$	**1**
(iii) $ax + ay + bx + by = a(x + y) + b(x + y) = (a + b)(x + y)$	**1**
(b)	
Common denominator 12 : $\frac{x}{3} + \frac{x + 1}{4} = \frac{4x}{12} + \frac{3(x + 1)}{12}$	**1**
$\frac{4x + 3x + 3}{12}$	**1**
$\frac{7x + 3}{12}$	**1**
(c)	
$(2x)^2 - 2(2x)(3) + 3^2$	**1**
$4x^2 - 12x + 9$	**1**

Total marks: 10

Question 3

(a)	
Multiplying both sides by 12: $4(x + 2) - 3(x - 1) = 24$	**1**
$4x + 8 - 3x + 3 = 24 \Rightarrow x + 11 = 24$	**1**
$x = 13$ (check: $5 - 3 = 2$)	**1**
(b)	
$2p + 3b = 340$ .....①	**1**
$4p + b = 280$ .....②	**1**
From ②: $b = 280 - 4p$; substituting in ①: $2p + 840 - 12p = 340 \Rightarrow -10p = -500 \Rightarrow p = 50$	**1**
$b = 280 - 200 = 80$; a pen costs Rs 50 and a book costs Rs 80 (check: $100 + 240 = 340$; $200 + 80 = 280$)	**1**
(c)	
$v = u + at \Rightarrow v - u = at$	**1**
$a = \frac{v - u}{t}$	**1**
$a = \frac{30 - 10}{4} = 5$	**1**

Total marks: 10

Question 4

(i)	
y = −3 when x = 0 ; y = 1 when x = 2 (both correct)	**1**
(ii)	
x- and y-axes drawn with a uniform, suitable scale (e.g. 2 cm = 1 unit on the x-axis ; 1 cm = 1 unit on the y-axis)	**1**
Points (−1, −5), (0, −3), (1, −1), (2, 1), (3, 3) plotted correctly	**1**
A straight line drawn through the points (it cuts the y-axis at −3 and the x-axis at 1.5)	**1**
(iii)	
Gradient = 2	**1**
Intercept = −3	**1**
(iv)	
Reading the point where the graph meets the x-axis (y = 0)	**1**
x = 1.5 (1.4 – 1.6 accepted ; check: 2(1.5) − 3 = 0)	**1**
(v)	
Parallel lines have equal gradients: m = 2	**1**
It passes through (0, 2), so the intercept is 2: y = 2x + 2	**1**

Total marks: 10

Question 5

(a)(i)	
$(5 - 2) \times 180°$	**1**
$= 540°$	**1**
(a)(ii)	
$x + 2x + 110° + 120° + 130° = 540° \Rightarrow 3x + 360° = 540°$	**1**
$3x = 180° \Rightarrow x = 60°$	**1**
(a)(iii)	
Number of sides $= \frac{360°}{40°}$	**1**
$= 9$	**1**
(b)	
Since AB = AC, $A\hat{B}C = A\hat{C}B$ (base angles of an isosceles triangle are equal)	**1**
$A\hat{B}C = (180° - 40°) \div 2 = 70°$ (angle sum of a triangle is 180°)	**1**
$x = 180° - 70°$ (angles on a straight line add up to 180°)	**1**
$x = 110°$ (check: exterior angle = 40° + 70°)	**1**

Total marks: 10

Question 6

(a)	
By Pythagoras theorem: $\text{height}^2 = 13^2 - 5^2$	**1**
$= 169 - 25 = 144$	**1**
Height = 12 m	**1**
(b)(i)	
$2\pi r = 2 \times \frac{22}{7} \times 35$	**1**
= 220 cm	**1**
(b)(ii)	
1.1 km = 1100 m = 110 000 cm	**1**
Number of turns = $\frac{110\,000}{220} = 500$	**1**
(c)	
Length of semicircular arc = $\frac{1}{2} \times 2 \times \frac{22}{7} \times 7 = 22$ cm	**1**
Diameter = 14 cm	**1**
Perimeter = 22 + 14 = 36 cm	**1**

Total marks: 10

Question 7

(i)	
The mark with the highest frequency (6): mode = 6	**1**
(ii)	
With 20 students the median is the mean of the 10th and 11th values ; cumulative frequencies 2, 7, 13 ⇒ both are 6	**1**
Median = 6	**1**
(iii)	
$\Sigma fx = 4\times2 + 5\times5 + 6\times6 + 7\times4 + 8\times3 = 8 + 25 + 36 + 28 + 24$	**1**
$= 121$	**1**
Mean = $\frac{121}{20} = 6.05$	**1**
(iv)	
Range = $8 - 4 = 4$	**1**
(v)	
Fraction of students who scored 6 = $\frac{6}{20}$	**1**
Angle of the sector = $\frac{6}{20} \times 360°$	**1**
$= 108°$	**1**

Total marks: 10